

## 1) Recap on P.T.

Events  $\subseteq$  Outcome  $\subseteq \Omega$  (PT space)

$$P[\emptyset] \leq P[A] \leq P[\Omega] = 1$$

### Probability Union

$$P[A \cup B] = P[A] + P[B] - P[A \cap B]$$

### Disjoint:

$$A_i \cap A_j = \emptyset \quad \forall i \neq j : \bigcup_i A_i = \Omega$$

### Law of total probability:

$$P[B] = \sum_i P[A_i \cap B]$$

### Independence:

$$P[A \cap B] = P[A]P[B] \implies A, B \text{ independent}$$

### Cumulative Distribution Function (CDF):

$$F(x) = P[X \leq x] = \int_{-\infty}^x p_X(x) dx$$

- $F(x) \in [0, 1]$
- Non decreasing
- *Right* continuous

### Expectation:

$$\mathbb{E}[g(X)] = \sum_i g(x_i) P[x_i]$$

spooky scary stuff (this notation means sum in discrete case and int in continuous)

$$\mathbb{E}[g(x)] = \int_{-\infty}^{\infty} g(x) dF(x)$$

### Moments:

$$\mathbb{E}[X^m] = \sum_i x_i^m P[x_i]$$

- $m = 1$  is the mean
  - $m = 2$  is the power
- Central moment is adjusted for mean, with  $m = 2$  it is the variance

$$\mathbb{E}[(X - \mu)^2] = \text{var}(X)$$

### Joint distribution and marginals:

$$F(x; y) = P[X \leq x; Y \leq y]$$

$$f_X(x) = \int f(x, y) dy \quad f_Y(y) = \int f(x, y) dx$$

if

$$F(x; y) = F[x]F[y] \implies iid$$

if

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] \implies \text{uncorrelated}$$

equivalence between iid and uncorrelated holds for gaussian rvs

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

**Conditional Probability:**

$$P[A|B] = \frac{P[A \cap B]}{P[B]} \text{ for } P[B] \neq 0$$

$$P[A \cap B] = P[A|B]P[B]$$

$$P[A] = \sum_i P[A \cap B_i] = \sum_i P[A|B_i]P[B_i]$$

$$\mathbb{E}[Y] = \sum \mathbb{E}[Y|x_i]P[x_i]$$

**Fourier Tx Characteristic Function:**

$$\phi(t) = \int e^{it\lambda} dF(\lambda) = \mathbb{E}[e^{itx}]$$

$$\frac{d\phi}{dt} = \mathbb{E}[ixe^{itx}] \stackrel{t=0}{=} it[x]$$

$$\frac{d^2\phi}{dt^2} = \mathbb{E}[(ix)^2 e^{itx}] \stackrel{t=0}{=} i^2 t[x^2]$$

**Generating functions:**

$$g(s) = \sum_{k=0}^{\infty} p_k s^k = \mathbb{E}[s^x] \stackrel{s=e^{it}}{=} \mathbb{E}[(e^{it})^x]$$

$$g'(s) = \mathbb{E}[xs^{x-1}] \stackrel{s=1}{=} \mathbb{E}[x]$$

$$g''(s) = \mathbb{E}[x(x-1)s^{x-2}] \stackrel{s=1}{=} \mathbb{E}[x^2] - \mathbb{E}[x]$$

then

$$\mathbb{E}[x^2] = g''(1) + g'(1)$$

## 1.1) Distributions

**Bernoulli:**

$$x \in \{0, 1\}$$

$$p = P[x = 1] = 1 - P[x = 0]$$

$$\mathbb{E}[x] = p \quad \mathbb{E}[x^2] = p \quad \text{var}(x) = p(p-1)$$

**Binomial:**

sum of N iid Bernoulli rv

$$y = \sum_{i=1}^n x_i \quad x_i \sim \text{bern}(p)$$

$$\mathbb{E}[y] = n\mathbb{E}[x] = np \quad \text{var}(y) = n\text{var}(x) = np(1-p)$$

with  $n \rightarrow \infty$  binomial becomes poisson with  $\lambda = np$

**Poisson:**

$$P[x = k] = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k \geq 0$$

$$\mathbb{E}[x] = \sum_{k=0}^{\infty} k e^{-\lambda} \frac{\lambda^k}{k!} = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{-\lambda} e^{\lambda} = \lambda$$

$$\text{var}(x) = \lambda$$

**Geometric RV:**

how many failures before the first success (0,0,0,1 will be 3 not 4)  $Z$

second definition: how many attempts until first success (definition of before +1)  $Z + 1$

$$P[Z = k] = (1 - p)^k p$$

$$\mathbb{E}[Z] = \sum_{k=0}^{\infty} k P[Z = k] = \sum_{k=0}^{\infty} P[k > Z] = \frac{1-p}{p} \quad \text{var}(Z) = \frac{1-p}{p^2}$$

**Exponential:**

$$f_T(t) = \begin{cases} \lambda e^{-\lambda t} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

$$\mathbb{E}[T] = \frac{1}{\lambda} \quad \text{var}(T) = \frac{1}{\lambda^2}$$

Memoryless property:

$$P[T - t > x | T > t] = \frac{P[T - t > x, T > t]}{P[T > t]} = \frac{P[T - t > x]}{P[T > t]} = \frac{e^{-\lambda(t+x)}}{e^{-\lambda t}} = e^{-\lambda x}$$

since  $T - t > x$  implies  $T > t + x$  and thus  $T > t$

**Gamma Distribution:**

$$\Gamma(\alpha) = (\alpha - 1)!$$

$$f(x) = \frac{\lambda}{\Gamma(\alpha)} (\lambda x)^{\alpha-1} e^{-\lambda x}$$

## 2) Important Stuff

For discrete time rvs we have

$$\mathbb{E}[X] = \sum_{n=1}^{\infty} nP[X = n] = \sum_{n=0}^{\infty} P[X > n]$$

### 3) Markov Chains

A Markov process is a stochastic process with the **markovian property**, that is,

$$P_{i,j}^{n,n+1} = P[x_{n+1} = j | x_0 = i_0, \dots, x_n = i_n] = P[x_{n+1} = j | x_n = i_n]$$

where  $i, j$  is the direction (from state  $i$  to state  $j$ ) at iteration  $n$  to  $n + 1$ .

Unless explicitly stated,  $P_{ij}^{n,n+1} = P_{ij}$

We can also define the **transition probability matrix**

$$P = \begin{bmatrix} P_{00} & P_{01} & \dots \\ P_{10} & P_{11} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

If  $P$  is finite, then these properties hold:

- $P_{ij} \in [0, 1]$
- $\sum_j P_{ij} = 1 \quad \forall i \text{ (rows)}$

A markov process is defined where  $P$  is completely known and also  $X_0$ .

If a specific series of events is required, it can be rewritten as:

$$P[x_0 = i_0, x_1 = i_1, \dots, x_n = i_n] = p_{i_0} P_{i_0, i_1} P_{i_1, i_2} \dots P_{i_{n-1}, i_n} = p_{i_0} \prod_{k=1}^n P_{i_{k-1}, i_k}$$

The probability of doing a transition in exactly  $n$  steps is:

$$P_{ij}^{(n)} = P[x_n = i_n | x_0 = i_0] = \sum_k P_{ik} P_{kj}^{(n-1)}$$

In matrix form:

$$P^{(n)} = P \times P \times \dots \times P = P^n$$

Proof of series of transitions:

By def of conditional probability and then markov process:

$$\begin{aligned} P[x_0 = i_0, \dots, x_n = i_n] &= P[x_0 = i_0, \dots, x_{n-1} = i_{n-1}] \cdot P[x_n = i_n | x_0 = i_0, \dots, x_{n-1} = i_{n-1}] \\ &= P[x_0 = i_0, \dots, x_{n-1} = i_{n-1}] \cdot P[x_n = i_n | x_{n-1} = i_{n-1}] \\ &= P[x_0 = i_0, \dots, x_{n-1} = i_{n-1}] \cdot P_{i_{n-1}, i_n} \end{aligned}$$

repeat  $n - 1$  times

$$= p_{i_0} \prod_{k=1}^n P_{i_{k-1}, i_k}$$

□

#### 3.1) First Step Analysis

First step analysis is a method to solve the markov process:

- Find possible results at the first step
- Invoke markov property and law of total probability

Let the states be  $0, \dots, N$ , then:

- Let the first  $r$  states  $(0, \dots, r-1)$  be transient:  $P_{ij} \rightarrow 0$  for  $i, j \in [0, r]$
- The other states  $r, \dots, N$  are absorbing:  $P_{ii} = 1$  for  $i \in [r, N]$

We can write the transition matrix as follows:

$$P = \begin{bmatrix} Q & R \\ 0 & I \end{bmatrix}$$

We want to find the following:

The first step analysis consists in finding the possible transitions from any state  $i$  to a set state  $k$ , that is:

$$u_i = P[X_T = k | X_0 = i] \quad \forall i$$

these can be found with the following formula:

$$u_i = P_{ik} + \sum_{j=0}^{r-1} P_{ij} u_j$$

It is also possible to find the avg time spent before being absorbed:

$$T = \min \{n \geq 0; X_n \geq r\}$$

the avg time that is spent starting in state  $i$  before being absorbed in state  $k$  can be found with the following formula:

$$w_i = g(i) + \sum_{j=0}^{r-1} P_{ij} w_j \quad g(i) = \begin{cases} 1 & \text{if transient} \\ 0 & \end{cases}$$

to find the avg time spent in a specific state  $j$  the formula is similar:

$$v_i = g(i) + \sum_{j=0}^{r-1} P_{ij} v_j \quad g(i) = \begin{cases} 1 & i = j \\ 0 & \end{cases}$$

**Example:**

$$P = \left[ \begin{array}{c|cc} \beta & \alpha & \gamma \\ \hline 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right]$$

Invoke the first step analysis and find probabilities to being absorbed to state  $k = 1$ :

$$u_i = P[X_T = 1 | X_0 = i] \rightarrow \begin{cases} u_0 = \alpha + \beta u_0 \\ u_1 = 1 + 0u_0 \\ u_2 = 0 + 0u_0 \end{cases} \rightarrow \begin{cases} u_0 = \frac{\alpha}{1-\beta} \\ u_1 = 1 \\ u_2 = 0 \end{cases}$$

if the starting probabilities are known, then the total probability of being absorbed to state 1 is

$$P[\text{absorbed in 1}] = \sum_{k=0}^2 p_k u_k$$

Now the times become:

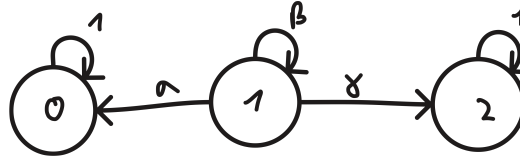
$$\begin{cases} w_0 = 1 + \beta w_0 \\ w_1 = 0w_0 \\ w_2 = 0w_0 \end{cases} \rightarrow \begin{cases} w_0 = \frac{1}{1-\beta} \\ w_1 = 0 \\ w_2 = 0 \end{cases}$$

**Example** (3 state MC without general formula).

*Consider the following markov process:*

$$P = \begin{bmatrix} 1 & 0 & 0 \\ \alpha & \beta & \gamma \\ 0 & 0 & 1 \end{bmatrix}$$

where  $\alpha + \beta + \gamma = 1$  and  $\alpha, \beta, \gamma \geq 0$   
 this can be visualized as an absorbing MC:



Representation

Two questions arise:

1. Starting from 1, what is the probability of being absorbed in 0?
2. How much time (on average) does it take to get there?

Invoke first step analysis we first need to see the various transition probabilities:

$$P[X_T = 0 | X_1 = 0] = 1 \quad P[X_T = 0 | X_1 = 1] = u \quad P[X_T = 0 | X_1 = 2] = 0$$

Now we can find  $u$ :

$$\begin{aligned} u &= P[X_T = 0 | X_0 = 1] = \sum_{k=0}^2 P[X_T = 0 | X_0 = 1, X_1 = k] P[X_1 = k | X_0 = 1] \\ &= \sum_{k=0}^2 P[X_T = 0 | X_1 = k] P[X_1 = k | X_0 = 1] = \sum_{k=0}^2 P[X_T = 0 | X_1 = k] P_{1k} \\ &= 1 \cdot \alpha + u \cdot \beta + 0 \cdot \gamma \\ \rightarrow u &= \frac{\alpha}{1 - \beta} = \frac{\alpha}{\alpha + \gamma} \end{aligned}$$

To find  $v$  we first calculate:

$$\mathbb{E}[T | X_1 = 0] = 0 \quad \mathbb{E}[T | X_1 = 1] = v \quad \mathbb{E}[T | X_1 = 2] = 0$$

we notice that  $T \geq 1$  and it stops once one of the absorbing states is reached:

$$\begin{aligned} v &= \mathbb{E}[T | X_0 = 1] = 1 + \sum_{k=0}^2 P_{1k} \mathbb{E}[T | X_1 = k] \\ &= 1 + \alpha \cdot \beta \cdot v + \gamma \cdot 0 \\ &= 1 + \beta v \\ \rightarrow v &= \frac{1}{1 - \beta} = \frac{1}{\alpha + \gamma} \end{aligned}$$

Alternatively a direct calculation can be done:

Notice that the probability is a geometric rv and thus:

$$\mathbb{E}[T | X_0 = 1] = \sum_{k=0}^{\infty} P[T > k | X_0 = 1] = \sum_k \beta^k = \frac{1}{1 - \beta}$$

**Example** (4 state MC).

Similar as before, but now with 4 states where 2 of them are absorbing.

$$P = \begin{bmatrix} P_{00} & P_{01} & P_{02} & P_{03} \\ P_{10} & P_{11} & P_{12} & P_{13} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Now find using the general formula for  $k = 2$ :

$$u_i = P[X_T = 2 | X_0 = i] \rightarrow \begin{cases} u_0 = P_{02} + P_{00}u_0 + P_{01}u_1 \\ u_1 = P_{12} + P_{10}u_0 + P_{11}u_1 \\ u_2 = 1 \\ u_3 = 0 \end{cases}$$

Now the times become:

$$\begin{cases} w_0 = 1 + P_{00}w_0 + P_{01}w_1 \\ w_1 = 1 + P_{10}w_0 + P_{11}w_1 \\ w_2 = 0 \\ w_3 = 0 \end{cases}$$

### 3.2) Random Walks

A random walk is a set of states where the transition can only occur between adjacent states (or in the same state), that is:

$$P = \begin{bmatrix} r_0 & p_0 & 0 & 0 & \dots & 0 & \dots \\ q_1 & r_1 & p_1 & 0 & \dots & 0 & \dots \\ 0 & q_2 & r_2 & p_2 & \dots & 0 & \dots \\ \vdots & \vdots & \ddots & \ddots & \ddots & & \\ 0 & & & q_i & r_i & p_i & \\ \vdots & & & & & \ddots & \dots \end{bmatrix}$$

$$P_{i,i+1} = p_i \quad P_{i,i-1} = q_i \quad P_{ii} = r_i$$

## 4) Proofs to know

### 4.1) Proof 1: Long-run Arrival Rate

For a renewal process, state precisely (also providing a formal proof) what is the value of:

$$\lim_{t \rightarrow \infty} \frac{N(t)}{t}$$

This is the **strong law of large numbers for renewal processes**

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Answer:

$$\lim_{t \rightarrow \infty} \frac{N(t)}{t} = \frac{1}{\mu} \text{ with probability 1}$$

Proof:

Let  $\{X_n\}_{n=1}^{\infty}$  be a sequence of iid rvs representing inter-arrival times between successive events with  $\mathbb{E}[X_i] = \mu \in (0, \infty)$

Let the arrival time be  $S_n = \sum_{i=1}^n X_i$  and the renewal counting process  $N(t)$  count the number of renewals up to time  $t$ , then

$$N(t) \geq n \iff S_n \leq t$$

Let the renewal  $N(t)$  occur at or before  $t$  by definition, then renewal  $N(t) + 1$  occurs strictly after  $t$ :

$$S_{N(t)} \leq t < S_{N(t)+1}$$

For  $N(t) \geq 1$  we have

$$\frac{S_{N(t)}}{N(t)} \leq \frac{t}{N(t)} < \frac{S_{N(t)+1}}{N(t)}$$

To evaluate these bounds, recall the strong law of large numbers:

$$\lim_{n \rightarrow \infty} \frac{S_n}{n} = \mu \text{ w.p. 1}$$

Since  $P[\lim_{t \rightarrow \infty} N(t) = \infty] = 1$  it is possible to asymptotically substitute  $n \rightarrow N(t)$  and get

$$\lim_{t \rightarrow \infty} \frac{S_{N(t)}}{N(t)} = \mu$$

For the upper bound we have

$$\lim_{t \rightarrow \infty} \frac{S_{N(t)+1}}{N(t)} = \lim_{t \rightarrow \infty} \frac{S_{N(t)+1}}{N(t)+1} \cdot \frac{N(t)+1}{N(t)} = \mu \cdot 1 = \mu$$

And thus by the squeeze theorem (two policemen) we have

$$\lim_{t \rightarrow \infty} \frac{t}{N(t)} = \mu \text{ w.p. 1} \implies \lim_{t \rightarrow \infty} \frac{N(t)}{t} = \frac{1}{\mu}$$

□

### 4.2) Proof 2: Recurrence is a Class Property

Prove that if states  $i$  and  $j$  of a Markov chain communicate and  $i$  is recurrent, then  $j$  is also recurrent.

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Proof:

Since  $i, j$  communicate ( $i \rightarrow j, j \rightarrow i \implies i \leftrightarrow j$ ), we have by definition

$$\exists m, n \geq 1 \text{ s.t. } P_{ji}^{(n)} > 0, P_{ij}^{(m)} > 0$$

And by definition of recurrent state (the probability that the chain eventually returns to state  $i$  in a finite number of steps is 1) we have

$$\sum_{k=0}^{\infty} P_{ii}^{(k)} = \infty$$

To prove that  $j$  is recurrent, we need to show that  $\sum_{k=0}^{\infty} P_{jj}^{(k)} = \infty$ .

Consider the following path: from  $j$  to  $i$  in  $n$  steps, then goes to  $i$  back to  $i$  in  $k$  steps and then goes back from  $i$  to  $j$  in  $m$  steps

$$P_{jj}^{(n+k+m)} = \sum_{r,s} P_{jr}^{(n)} P_{rs}^{(k)} P_{sj}^{(m)} \geq P_{ji}^{(n)} P_{ii}^{(k)} P_{ij}^{(m)}$$

Now sum along all  $k$ :

$$\sum_{k=0}^{\infty} P_{jj}^{(n+k+m)} \geq \sum_{k=0}^{\infty} P_{ji}^{(n)} P_{ii}^{(k)} P_{ij}^{(m)} = P_{ji}^{(n)} P_{ij}^{(m)} \sum_{k=0}^{\infty} P_{ii}^{(k)} = \infty$$

and thus  $j$  is recurrent

□

### 4.3) Proof 3: Proof of i.i.d. Exponential Inter-Arrival Times for Poisson Processes

For a Poisson process of rate  $\lambda$ , prove that the inter-arrival times are iid exponential with mean  $1/\lambda$ .

Proof:

By definition of a Poisson process, we have

- $N(0) = 0$
- Independent increments: for any  $0 \leq t_1 < \dots < t_k$  the rvs  $N(t_i) - N(t_{i-1})$  are independent
- Stationary Poisson increments: for any  $s \geq 0$  and  $t > 0$ , the increment  $N(s+t) - N(s) \sim \text{Poisson}(\lambda t)$ , meaning:

$$\mathbb{P}(N(s+t) - N(s) = k) = P(N(t) = k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}$$

For  $n = 1$ :

$$\mathbb{P}(T_1 > t_1) = \mathbb{P}(N(t_1) = 0) = \frac{(\lambda t_1)^0 e^{-\lambda t_1}}{0!} = e^{-\lambda t_1} \implies T_1 \sim \text{Exp}(\lambda)$$

For  $n \geq 2$ , condition on the arrival history ( $T_1 = t_1, \dots, T_{n-1} = t_{n-1}$ ) and let  $S_{n-1} = \sum_{i=1}^{n-1} t_i$ :

$$\begin{aligned} \mathbb{P}(T_n > t_n \mid T_1 = t_1, \dots, T_{n-1} = t_{n-1}) &= \mathbb{P}(N(S_{n-1} + t_n) - N(S_{n-1}) = 0 \mid T_1 = t_1, \dots, T_{n-1} = t_{n-1}) \\ &= \mathbb{P}(N(S_{n-1} + t_n) - N(S_{n-1}) = 0) \quad (\text{by independent increments}) \\ &= \mathbb{P}(N(t_n) = 0) \quad (\text{by stationary increments}) \\ &= e^{-\lambda t_n} \end{aligned}$$

Since this conditional probability is constant with respect to  $(t_1, \dots, t_{n-1})$ ,  $T_n$  is independent of  $(T_1, \dots, T_{n-1})$  with unconditional distribution  $T_n \sim \text{Exp}(\lambda)$ .

By induction,  $\{T_n\}_{n=1}^{\infty}$  are mutually independent and identically distributed  $\text{Exp}(\lambda)$ .

The mean of an exponential distribution  $T$  with rate  $\lambda$  is given by

$$\mathbb{E}[T] = \int_0^{\infty} P[T > t] dt = \int_0^{\infty} e^{-\lambda t} dt = -\frac{e^{-\lambda t}}{\lambda} \Big|_0^{\infty} = \frac{1}{\lambda}$$

□

#### 4.4) Proof 4: Inference from Past to Future and Vice Versa

For a Poisson process  $X(t)$  of rate  $\lambda$ , state and derive the expression of  $P[X(u) = k | X(t) = n]$  for the two cases:

- $0 < u < t, 0 \leq k \leq n$
- $0 < t < u, 0 \leq n \leq k$

Answer:

- $X(u) | X(t) = n \sim \text{Binomial}(n, \frac{u}{t}) \rightarrow P[\dots] = \binom{n}{k} \left(\frac{u}{t}\right)^k \left(1 - \frac{u}{t}\right)^{n-k}$
- $X(u) | X(t) = n \sim \text{Poisson}(\lambda(u-t)) \rightarrow P[\dots] = \frac{(\lambda(u-t))^{k-n} e^{-\lambda(u-t)}}{(k-n)!}$

Proof:

From the conditional probability definition we have:

$$P[X(u) = k | X(t) = n] = \frac{P[X(u) = k, X(t) = n]}{P[X(t) = n]}$$

We will use this as a baseline for our proofs

Notice that the joint probability can be rewritten as two disjoint events, depending on the case, either  $[0, u]$ ,  $[u, t]$  or  $[0, t]$ ,  $[t, u]$ .

Case 1: Infer past from future knowledge

We can rewrite the joint probability as two disjoint events  $[0, u]$ ,  $[u, t]$ :

$$P[X(u) = k, X(t) = n] = P[X(u) = k] \cdot P[X(t) - X(u) = n - k]$$

The first term is directly given by the Poisson distribution:

$$P[X(u) = k] = \frac{(\lambda u)^k e^{-\lambda u}}{k!}$$

While the second term is given from the stationary increments:

$$P[X(t) - X(u) = n - k] = P[X(t - u) = n - k] = \frac{(\lambda(t - u))^{n-k} e^{-\lambda(t-u)}}{(n-k)!}$$

Now

$$\begin{aligned}
P[X(u) = k | X(t) = n] &= \frac{(\lambda u)^k e^{-\lambda u}}{k!} \frac{(\lambda(t-u))^{n-k} e^{-\lambda(t-u)}}{(n-k)!} \frac{n!}{(\lambda t)^n e^{-\lambda t}} \\
&= \frac{n!}{k!(n-k)!} \frac{\lambda^k u^k \lambda^{n-k} (t-u)^{n-k}}{\lambda^n t^n} \frac{e^{-\lambda(u+t-u)}}{e^{-\lambda t}} \\
&= \binom{n}{k} \frac{u^k (t-u)^{n-k}}{t^n} \\
&= \binom{n}{k} \frac{u^k}{t^k} \frac{(t-u)^{n-k}}{t^{n-k}} \\
&= \binom{n}{k} \left(\frac{u}{t}\right)^k \left(1 - \frac{u}{t}\right)^{n-k}
\end{aligned}$$

Case 2: Predict future from past observations

We can rewrite the joint probability as two disjoint events  $[0, t]$ ,  $[t, u]$ :

$$P[X(u) = k, X(t) = n] = P[X(t) = n] \cdot P[X(u) - X(t) = k - n]$$

Notice that the first term is the denominator of the expression of the conditional probability, so these two cancel out

$$P[X(u) = k | X(t) = n] = P[X(u) - X(t) = k - n]$$

As before from the stationary increments we have that this is distributed as a Poisson random variable

$$X(u) - X(t) \sim \text{Poisson}(\lambda(u-t)) \rightarrow P[X(u) = k | X(t) = n] = \frac{(\lambda(u-t))^{k-n} e^{-\lambda(u-t)}}{(k-n)!}$$

□

#### 4.5) Proof 5: Period is a Class Property

State precisely and prove formally the result that establishes that in a Markov chain the period is a class property

For study purposes I include also the definition of period:

$$\delta(i) = \gcd \left\{ n \geq 1 : P_{ii}^{(n)} > 0 \right\}$$

if it is 1 it is aperiodic.

Proposition:

Let  $\{X_k\}_{k \geq 0}$  be a discrete-time Markov chain on a countable state space  $S$  with transition matrix  $P$ . For any state  $i \in S$ , the period  $d(i)$  (or  $\delta(i)$ ) is defined as:

$$d(i) = \gcd \{ n \geq 1 : P_{ii}^{(n)} > 0 \}$$

If two states  $i, j \in S$  communicate ( $i \leftrightarrow j$ ), then  $d(i) = d(j)$ .

Proof:

Since  $i \leftrightarrow j$  then  $\exists m, n : P_{ij}^{(m)} > 0, P_{ji}^{(n)} > 0$  and therefore

$$P_{jj}^{(m+n)} \geq P_{ji}^{(n)} P_{ij}^{(m)} > 0$$

and clearly  $\delta(j)$  must divide  $m + n$ :

$$\delta(j) | (m + n)$$

Let  $s \in \left\{ s \geq 1 : P_{ii}^{(s)} > 0 \right\}$ , that is, any step length such that  $P_{ii}^{(s)} > 0$ , then

$$P_{jj}^{(m+s+n)} \geq P_{ji}^{(n)} P_{ii}^{(s)} P_{ij}^{(m)} > 0$$

now it is clear that  $\delta(j)$  must also divide  $m + s + n$ :

$$\delta(j) | (m + s + n)$$

Now focus on  $\delta(j)$ . If it divides  $m + n$  and also  $m + s + n$  it must divide their difference, that is  $\delta(j)$  also divides  $s$ .

$$\delta(j) | ((m + s + n) - (m + n)) = s \rightarrow \delta(j) | s$$

Then,  $\delta(j)$  is a common divisor of the set  $\{s \geq 1 : P_{ii}^{(s)} > 0\}$  whose greatest common divisor is  $\delta(i)$  it yields that

$$\delta(j) \leq \delta(i)$$

By symmetry the roles of  $i$  and  $j$  can be reversed and the result becomes  $\delta(i) \leq \delta(j)$

By combining both inequalities we get

$$\begin{cases} \delta(j) \leq \delta(i) \\ \delta(i) \leq \delta(j) \end{cases} \implies \delta(j) = \delta(i)$$

□

#### 4.6 ! Proof 6: Derivation of $\mathbb{E}[S_{N(t)+1}]$ Using Renewal Theory

For renewal processes, give an expression for  $\mathbb{E}[S_{N(t)+1}]$  also providing a formal proof

Answer:

$$\mathbb{E}[S_{N(t)+1}] = \mathbb{E}[X](M(t) + 1)$$

where  $M(t)$  is the renewal function.

Proof 1:

Let:

- $(X_n)_{n \geq 1}$  be an i.i.d. sequence of positive inter-arrival times with mean  $E[X_i] = \mu \in (0, \infty)$
- $S_n = \sum_{i=1}^n X_i$  with  $S_0 = 0$
- $M(t) = \mathbb{E}[N(t)]$

Define  $A(t) = \mathbb{E}[S_{N(t)+1}]$ . Then:

$$\begin{aligned}
\mathbb{E}[S_{N(t)+1}] &= \mathbb{E}\left[\sum_{i=1}^{N(t)+1} X_i\right] \\
&= \mathbb{E}\left[\sum_{i=1}^{\infty} X_i \mathbf{1}_{\{N(t)+1 \geq i\}}\right] \\
&= \sum_{i=1}^{\infty} \mathbb{E}[X_i \mathbf{1}_{\{N(t)+1 \geq i\}}] \quad (\text{by Tonelli's Theorem, terms are non-negative}) \\
&= \sum_{i=1}^{\infty} \mathbb{E}[X_i] \mathbb{E}[\mathbf{1}_{\{N(t)+1 \geq i\}}] \quad (\text{since } \{N(t) + 1 \geq i\} = \{S_{i-1} \leq t\} \text{ is independent of } X_i) \\
&= \sum_{i=1}^{\infty} \mathbb{E}[X_i] \mathbb{P}(N(t) + 1 \geq i) \\
&= \mathbb{E}[X_1] \sum_{i=1}^{\infty} \mathbb{P}(N(t) + 1 \geq i) \quad (\text{since } X_i \text{ are identically distributed}) \\
&= \mathbb{E}[X_1] \mathbb{E}[N(t) + 1] \quad (\text{tail-sum formula for non-negative integer RVs}) \\
&= \mathbb{E}[X_1] (M(t) + 1)
\end{aligned}$$

□

Proof 2:

Using the renewal argument (first step analysis) we have

$$\mathbb{E}[S_{N(t)+1} | X_1 = x] = \begin{cases} x & \text{if } x > t \text{ (no renewal yet)} \\ x + A(t - x) & \text{if } x \leq t \text{ (add time from } x \text{ to arrival)} \end{cases}$$

We can apply the law of total expectation:

$$\begin{aligned}
\mathbb{E}[S_{N(t)+1}] &= A(t) = \int_0^{\infty} \mathbb{E}[S_{N(t)+1} | X_1 = x] dF(x) \\
&= \int_0^t x + A(t - x) dF(x) + \int_t^{\infty} x dF(x) \\
&= \int_0^{\infty} x dF(x) + \int_0^t A(t - x) dF(x) \\
&= \mathbb{E}[X] + \int_0^t A(t - x) dF(x)
\end{aligned}$$

This is a standard renewal integral equation of the form:

$$A(t) = a(t) + \int_0^t A(t - x) dF(x), \quad t \geq 0$$

where  $a(t) = \mathbb{E}[X]$  is constant with respect to  $t$ .

By the general theorem for renewal equations, the unique bounded solution is given by:

$$A(t) = a(t) + \int_0^t a(t - x) dM(x) = \mathbb{E}[X_1] + \int_0^t \mathbb{E}[X_1] dM(x) = \mathbb{E}[X_1] + \mathbb{E}[X_1] M(t)$$

where  $a(t - x) = \mathbb{E}[X_1]$  and since it is constant it is removed from the integral.

□

## 4.7) Proof 7: Rehash of Proof 6

See [Proof 6](#).

## 4.8) Proof 8: Elementary Renewal Theorem TODO

State and Prove the Elementary Renewal Theorem

Let  $\{N(t), t \geq 0\}$  be a renewal process with independent and identically distributed inter-arrival times  $\{X_n\}_{n \geq 1}$  having mean  $\mu = \mathbb{E}[X_1]$ .

Let  $M(t) = \mathbb{E}[N(t)]$  be the renewal function. Then:

$$\lim_{t \rightarrow \infty} \frac{M(t)}{t} = \frac{1}{\mu}$$

(where  $1/\mu$  is interpreted as 0 if  $\mu = \infty$ ).

Proof:

By definition of  $N(t)$ , the  $N(t)$ -th renewal occurs on or before  $t$ , and the  $(N(t) + 1)$ -th renewal occurs strictly after  $t$ :

$$S_{N(t)} \leq t < S_{N(t)+1}$$

Dividing through by  $N(t)$  for  $t$  large enough such that  $N(t) \geq 1$ :

$$\frac{S_{N(t)}}{N(t)} \leq \frac{t}{N(t)} < \frac{S_{N(t)+1}}{N(t)+1} \cdot \frac{N(t)+1}{N(t)}$$

By the SLLN,  $\lim_{n \rightarrow \infty} \frac{S_n}{n} = \mu$  therefore

$$\lim_{t \rightarrow \infty} \frac{N(t)}{t} = \frac{1}{\mu} \quad \text{a.s.}$$

Fatou's Lemma can be directly applied to find the lower bound:

$$\liminf_{t \rightarrow \infty} \frac{\mathbb{E}[N(t)]}{t} = \liminf_{t \rightarrow \infty} \mathbb{E} \left[ \frac{N(t)}{t} \right] \geq \mathbb{E} \left[ \liminf_{t \rightarrow \infty} \frac{N(t)}{t} \right] = \frac{1}{\mu}$$

#### 4.9) Proof 9: Rehash of Proof 5

Prove that in a Markov Chain the period is a class property, see [Proof 5](#).

#### 4.10) Proof 10: Random Walk Analysis TODO IMPORTANT

Consider a random walk over non-negative integers with the following transition probabilities:

$$P_{01} = 1, \quad P_{i,i+1} = p, \quad P_{i,i-1} = q, \quad i > 0$$

with  $p + q = 1$

Study its behavior, in particular characterize its recurrence or transience and derive the steady state distribution.

Let's start from the classification.

The chain is **irreducible** since all states are communicating, so all states share the same classification.

Moreover the **period is 2** since it requires always an even number of steps to return to state  $i$ .

Now study recurrence/transience:

Let  $h_i$  denote the probability of eventually reaching state 0 from state  $i$ . Clearly  $h_0 = 1$

$$h_i = ph_{i+1} + qh_{i-1}$$

Multiply lhs by  $p + q = 1$

$$\begin{aligned} ph_i + qh_i &= ph_{i+1} + qh_{i-1} \\ p(h_i - h_{i+1}) &= q(h_{i-1} - h_i) \end{aligned}$$

$$\Delta_{i+1} = h_i - h_{i+1} = \frac{q}{p}(h_{i-1} - h_i) = \frac{q}{p}\Delta_i$$

Let's see how  $\Delta_i$  behaves

$$\begin{aligned} \Delta_1 &= h_0 - h_1 = 1 - h_1 \\ \Delta_2 &= \frac{q}{p}\Delta_1 \\ \Delta_3 &= \frac{q}{p}\Delta_2 = \left(\frac{q}{p}\right)^2 \Delta_1 \\ &\dots \\ \Delta_i &= \left(\frac{q}{p}\right)^{i-1} \Delta_1 \end{aligned}$$

and thus

$$h_0 - h_n = (h_0 - \underbrace{h_1}_{=0}) + (h_1 - h_2) + \dots + (h_{n-1} - h_n) = \sum_{i=1}^n \Delta_i = \Delta_1 \sum_{k=0}^{n-1} \left(\frac{q}{p}\right)^k$$

Since  $h_0 = 1$  is known we have

$$h_n = 1 - \Delta_1 \sum_{k=0}^{n-1} \left(\frac{q}{p}\right)^k$$

This has 2 cases:

- The series diverges for  $p \leq q$ , then  $\lim_{n \rightarrow \infty} h_n = -\infty$  which violates  $h_n \geq 0$ . To fix this then  $h_1$  must be equal to 1 so to have  $\Delta_1 = 0$  and  $h_n = 1$  and the chain is recurrent
- The series converges for  $p > q$  and thus  $\lim_{n \rightarrow \infty} h_n = 1 - \Delta_1 \frac{p}{p-q}$ . Since every step is independent we also have  $h_n = (h_1)^n \rightarrow \lim_{n \rightarrow \infty} h_n = 0$ . Now it is clear that  $0 = 1 - (1 - h_1) \frac{p}{p-q} \rightarrow h_1 = \frac{q}{p}$  and thus  $h_n = \left(\frac{q}{p}\right)^n$  and the chain is transient

The steady state distribution must only be studied in the recurrent case, so for  $p \leq q \rightarrow p < 1/2$ .  
To find the stationary distribution  $\pi = (\pi_0, \pi_1, \pi_2, \dots)$ , we solve the system:

$$\pi = \pi P \quad \text{subject to} \quad \sum_{n=0}^{\infty} \pi_n = 1 \quad \text{and} \quad \pi_n \geq 0$$

Which is

$$\pi_j = \sum \pi_i P_{ij}$$

State 0:

$$\pi_0 = \pi_1 P_{10} = q\pi_1 \rightarrow \pi_1 = \frac{1}{q}\pi_0$$

State 1:

$$\pi_1 = \pi_0 P_{01} + \pi_2 P_{21} = \pi_0 + q\pi_2 \rightarrow \pi_2 = \frac{p}{q^2}\pi_0$$

Generic state  $j$ :

$$\pi_j = p\pi_{j-1} + q\pi_{j+1} \rightarrow \pi_n = \frac{\pi_0}{q} \left(\frac{p}{q}\right)^{n-1}$$

Now normalize:

$$\pi_0 + \frac{\pi_0}{q} \sum_{i=0}^{\infty} \left(\frac{p}{q}\right)^i = 1 \rightarrow \pi_0 \left(\frac{q-p+1}{q-p}\right) = 1 \rightarrow \pi_0 = \frac{q-p}{2q}$$

And finally

$$\pi_n = \dots$$

#### 4.11) ! Proof 11: Finite MC existence conditions of states

Prove that a Markov chain with a finite number of states cannot have any null recurrent state

---

Proof:

A state  $i$  is null recurrent if it is recurrent ( $\mathbb{P}(T_i < \infty \mid X_0 = i) = 1$ ) and its expected return time is infinite:

$$\mu_i = \mathbb{E}[T_i \mid X_0 = i] = \infty$$

**By contradiction** suppose  $\exists i$  null recurrent:

**a)** Recurrence is a class property and since  $C$  (where  $i \in C$ ) is communicating and recurrent it must be closed. By total probability:

$$\sum_{j \in C} P_{ij}^{(n)} = 1, \quad \forall i \in C, \forall n \geq 1$$

**b)** A property of null recurrence classes is to have zero long-run transition probabilities:

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N P_{ij}^{(n)} = \frac{1}{\mu_j} = 0, \quad \forall i, j \in C$$

Now, find the fraction of time that the chain remains in  $C$ ; using property **a)**:

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \sum_{j \in C} P_{ij}^{(n)} = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N 1 = 1$$

Because the state space  $S$  is finite, the class  $C \subseteq S$  is finite ( $|C| < \infty$ ). Therefore, the finite sum and the limit commute. Applying **b)** yields:

$$\lim_{N \rightarrow \infty} \sum_{j \in C} \left( \frac{1}{N} \sum_{n=1}^N P_{ij}^{(n)} \right) = \sum_{j \in C} \left( \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N P_{ij}^{(n)} \right) = \sum_{j \in C} 0 = 0$$

which is a contradiction. □

##### 4.11.1) Proof Bonus:

Prove that in a MC with a finite number of states, there must be at least one recurrent state

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Proof:

We have already shown that in a finite MC there cannot be any null recurrent states.

Suppose by contradiction that there are no recurrent states. Then every state  $i \in C$  is transient.

**a)** A transient state has a finite expected number of visits, and therefore since this infinite series converges, its terms must tend to zero:

$$\sum_{n=1}^{\infty} P_{ij}^{(n)} < \infty \rightarrow \lim_{n \rightarrow \infty} P_{ij}^{(n)} = 0 \quad \forall i, j \in C$$

**b)** Because the chain is finite, it must be in some state at step  $n$ , so

$$\sum_{j \in C} P_{ij}^{(n)} = 1$$

Because the state space  $C$  is finite, the limit as  $n \rightarrow \infty$  commutes with the summation:

$$1 = \lim_{n \rightarrow \infty} \sum_{j \in C} P_{ij}^{(n)} = \sum_{j \in C} \left( \lim_{n \rightarrow \infty} P_{ij}^{(n)} \right) = 0$$

Therefore, there must exist at least one positive recurrent state. □

#### 4.12) Proof 12: Rehash of Proof 4, only infer past from future knowledge

Prove that for a Poisson process  $X(t)$  the statistics of  $X(s)$  conditioned on  $X(t)$ ,  $s < t$ , is binomial and provide the expression of  $P[X(s) = k | X(t) = n]$

Answer

By the total probability formula we have that:

$$P[X(s) = k | X(t) = n] = \frac{P[X(s) = k \cap X(t) = n]}{P[X(t) = n]}$$

Where  $P[X(t) = n] = (\lambda t)^n e^{-\lambda t} / n!$

The joint probability can be rewritten as two disjoint events:

$$P[X(s) = k \cap X(t) = n] = P[X(s) = k] \cdot P[X(t) - X(s) = n - k]$$

The first is straightforward as  $P[X(s) = k] = (\lambda s)^k e^{-\lambda s} / k!$

The second is distributed as Poisson( $\lambda(n - k)$ ) so  $P[X(t) - X(s) = n - k] = (\lambda(n - k))^{t-s} e^{-\lambda(n-k)} / (t - s)!$

Finally put everything together, do some algebraic manipulations and you get

$$X(s) | X(t) = n \sim \text{Binomial}(n, \frac{s}{t}) \quad p = \binom{n}{k} \left( \frac{s}{t} \right)^k \left( 1 - \frac{s}{t} \right)^{n-k}$$

□

#### 4.13) Proof 13: N-Step Probability Expression

Prove that for a Markov chain the n-step transition probabilities  $P_{ij}^{(n)}$  satisfy the relationship

$$P_{ij}^{(n)} = \sum_m P_{im}^{(k)} P_{mj}^{(n-k)}$$

Proof:

From the definition:

$$\begin{aligned}
P_{ij}^{(n)} &= P[X_n = j | X_0 = i] \\
&= \sum_m P[X_n = j, X_k = m | X_0 = i] \quad \text{with } 0 < k \leq n \\
&= \sum_m \frac{P[X_n = j, X_k = m, X_0 = i]}{P[X_0 = i]} \\
&= \sum_m \frac{P[X_0 = i] P[X_k = m | X_0 = i] P[X_n = j | X_k = m, X_0 = i]}{P[X_0 = i]} \\
&= \sum_m P[X_k = m | X_0 = i] P[X_n = j | X_k = m, X_0 = i] \\
&= \sum_m P_{im}^{(k)} P_{mj}^{(n-k)} \quad \text{recall markov property: } P[X_n = j | X_k = m, X_0 = i] = P[X_n = j | X_k = m]
\end{aligned}$$

□

#### 4.14 ! Proof 14: Define Recurrent States

Give the definition of a recurrent state and show that a state is recurrent if and only if

$$\sum_{n=1}^{\infty} P_{ii}^{(n)} = \infty$$

Definition:

A state  $i$  is **recurrent** if the probability of eventually returning to state  $i$ , given that the chain starts in  $i$ , is equal to 1 :

$$f_{ii} = \mathbb{P}(T_i < \infty | X_0 = i) = 1$$

where  $T_i = \min\{n \geq 1 : X_n = i\}$  is the first return time to state  $i$ .

Proof:

Let  $M_i$  be the total number of returns to state  $i$  over  $n$  steps:

$$M_i = \sum_{n=1}^{\infty} u_{X_n=i}$$

where  $u_{X_n=i}$  is a counter switch (1 if  $X_n = i$ )

First consider these two preliminary results:

a)

$$\mathbb{E}[u_{X_n=i}] = 1 \cdot P[X_n = i] + 0 \cdot P[X_n \neq i] = P[X_n = i]$$

$$\mathbb{E}[u_{X_n=i} | X_0 = i] = P[X_n = i | X_0 = i] \longrightarrow P_{ii}^{(n)}$$

b) For  $k = 1$  by definition:

$$\begin{aligned}
\mathbb{P}(M_i \geq 1 | X_0 = i) &= \mathbb{P}(\text{chain returns to } i \text{ at least 1 time} | X_0 = i) \\
&= \mathbb{P}(T_i^{(1)} < \infty | X_0 = i) \\
&= f_{ii}
\end{aligned}$$

General case by induction

$$\begin{aligned}
\mathbb{P}(M_i \geq k | X_0 = i) &= \mathbb{P}(M_i \geq k-1 | X_0 = i) \cdot \mathbb{P}(M_i \geq k | M_i \geq k-1, X_0 = i) \\
&= \mathbb{P}(M_i \geq k-1 | X_0 = i) \cdot \mathbb{P}(M_i \geq 1 | X_0 = i) \\
&= \mathbb{P}(M_i \geq k-1 | X_0 = i) \cdot f_{ii} \\
&= (f_{ii})^{k-1} \cdot f_{ii} \\
&= (f_{ii})^k
\end{aligned}$$

The expectance of  $M_i$  starting from  $i$  using a)

$$\mathbb{E}[M_i | X_0 = i] = \mathbb{E}\left[\sum_{n=1}^{\infty} u_{X_n=i} | X_0 = i\right] = \sum_{n=1}^{\infty} \mathbb{E}[u_{X_n=i} | X_0 = i] = \sum_{n=1}^{\infty} P_{ii}^{(n)}$$

But this can also be rewritten using the tail sum formula ( $\mathbb{E}[X] = \sum_{n=1}^{\infty} P(X \geq n)$ ) using b)

$$\mathbb{E}[M_i | X_0 = i] = \sum_{k=1}^{\infty} P[M_i \geq k | X_0 = i] = \sum_{k=1}^{\infty} (f_{ii})^k$$

These expressions are equal, so we have

$$\sum_{n=1}^{\infty} P_{ii}^{(n)} = \sum_{k=1}^{\infty} (f_{ii})^k = \infty$$

#### 4.15) Proof 15: Define Transient States

Give the definition of a transient state and show that a state is transient if and only if

$$\sum_{n=1}^{\infty} P_{ii}^{(n)} < \infty$$

Answer:

A state  $i$  is **transient** if the probability of ever returning to state  $i$ , given that the chain started in state  $i$ , is strictly less than 1:

$$f_{ii} = \mathbb{P}(T_i < \infty | X_0 = i) < 1$$

where  $T_i = \min\{n \geq 1 : X_n = i\}$  is the first return time to state  $i$ .

Equivalently, there is a strictly positive probability  $(1 - f_{ii}) > 0$  that the chain leaves state  $i$  and **never returns**.

Proof:

Same as before, but now since  $f_{ii} < 1$  we have

$$\sum_{n=1}^{\infty} P_{ii}^{(n)} = \sum_{k=1}^{\infty} (f_{ii})^k = \frac{f_{ii}}{1 - f_{ii}} < \infty$$

## 5) Exercises

### 5.1) Poisson Process

Definition of Poisson Process PP:

$$X(t) \sim \text{Poi}(\lambda t) \implies \begin{aligned} P[X(t) = k] &= \frac{(\lambda t)^k e^{-\lambda t}}{k!}, \quad k \geq 0 \\ \mathbb{E}[X(t)] &= \lambda t \\ \text{var}(X(t)) &= \lambda t \end{aligned}$$

Moreover

$$X(t_1) - X(t_2) \sim \text{Poi}((t_1 - t_2)\lambda), \quad t_1 > t_2$$

A fundamental property of a Poisson process is that the number of events occurring in non-overlapping intervals are independent.

$$P[A|B] = P[A] \iff A \perp B$$

Recall:

$$P[A|B] = \frac{P[A, B]}{P[B]}$$

In the exercises, this is used only when:

- one PP conditioned on one past or future event = One PP with max 2 sampled points

Total distribution is used

- One PP with at least 3 sampled points

This total probability formula is useful:

$$P(E | C) = \sum_{k \in \mathcal{K}} P(E | B_k \cap C) \cdot P(B_k | C)$$

Practical example:

$$\begin{aligned} P[X_1(3) = 2 | X_1(2) + X_2(2) = 1] &= \\ = \sum_{k=0}^1 P[X_1(3) = 2 | X_1(2) = k, X_1(2) + X_2(2) = 1] \cdot P[X_1(2) = k | X_1(2) + X_2(2) = 1] \end{aligned}$$

### 5.2) Poisson Process Queues

In a queue we have the arrival distribution  $A$  and service time  $S$ .

Usually we have  $A(t) \sim \text{Poi}(\lambda t)$  and  $S$  either uniform, constant or exponentially distributed.

From here we define the number of customers in the system at time  $t$  as  $N(t)$ .

If there are infinite queues/capacity, then  $N(t) \sim \text{Poi}(\lambda_p(t))$  with rate

$$\lambda_p(t) = \lambda \int_0^t 1 - G(z) dz \xrightarrow{t \rightarrow \infty} \lambda E[S]$$

Where  $G(z)$  is the CDF of the Service time. Notice that  $t \rightarrow \infty$  doesn't need to be the limit, just  $t >$  then when the service time CDF converges to 1.

But notice that

$$P[N(t) = k] = \frac{[\lambda_p(t)]^k e^{-\lambda_p(t)}}{k!}$$

It is possible to find the probability of having  $n$  customers in service at  $t$  with  $k$  arrivals before using this:

- Constant service time  $\beta$ :

$$\begin{aligned} P[N(t) = n | A(t) = k] &= \frac{P[N(t) = n, A(t) = k]}{P[A(t) = k]} = \frac{P[A(t - \beta) = k - n] P[A(\beta) = n]}{P[A(t) = k]} \\ &= \binom{k}{n} \left(\frac{\beta}{t}\right)^n \left(\frac{t - \beta}{t}\right)^{k-n} \\ &\sim \text{Binomial}\left(k, \frac{\beta}{t}\right) \end{aligned}$$

- Random service time:

$$\begin{aligned} P[N(t) = n | A(t) = k] &= \binom{k}{n} \left[\frac{\lambda_p(t)}{\lambda t}\right]^n \left[1 - \frac{\lambda_p(t)}{\lambda t}\right]^{k-n}, \quad 0 \leq n \leq k \\ &\sim \text{Binomial}\left(k, \frac{\lambda_p(t)}{\lambda t}\right) \end{aligned}$$

Most importantly, the constant service time case is a special case of the random service time.

### 5.3) MC

#### Analysis of MC:

- Diagram
- Classes
- Recurrence/transience
- Periodicity

#### Probability absorption into class $a$ :

If the transient state is only one ( $|C_t| = 1$ ):  $P[\text{go to class } a] / P[\text{leave trans state}]$

If the transient state is a class with  $|C_t| > 1$  then first step analysis is required:

Define:  $\mu_i = P[\text{go to } a \text{ starting from } i]$

$$\mu_i = \sum_j P_{ij} \mu_j$$

Boundary condition:  $\mu_a = 1 \quad \mu_b = 0, \quad b \in \{\text{all other abs states}\}$

where the boundary condition is a direct result of the summation.

#### Compute Steady State Probabilities:

For each irreducible **positive** recurrent class solve:

$$\begin{cases} \pi_C P_C = \pi_C \\ \sum_{\pi_i \in \pi_C} \pi_i = 1 \end{cases}$$

For each irreducible transient or negative recurrent class it is 0.

#### Compute Avg Recurrence Times:

For each irreducible **positive** recurrent class solve:

$$m_i = \frac{1}{\pi_i}$$

For each irreducible transient or negative recurrent class it is  $\infty$ .

### Avg Number of Visits starting from $a$

Starting from  $a$  inside a pos rec chain:

$$W_{a\bullet}^{(n)} = \sum_{k=0}^{n-1} [P^k]_{a\bullet} \stackrel{n \rightarrow \infty}{\approx} n\pi_{\bullet}$$

Notice that this can be easily computed as:

Set

$$X_0 = (0, \dots, 0, \underbrace{1}_{a\text{-th pos}}, 0, \dots, 0)$$

Find  $X_k = X_{k-1}P$ ,  $k = 1, \dots, n-1$  (where  $X_1$  is the  $a$ -th row of  $P$  and  $X_0$  is a null row where only  $a$ -th entry is 1).

$$X_k = X_{k-1}P = X_0P^k = [P^k]_{a\bullet}$$

Then the  $W_{a\bullet}^{(n)}$  matrix is just the sum, to find  $j$  just select the  $j$ -th entry

$$W_{a\bullet}^{(n)} = \sum_{k=0}^{n-1} X_k \stackrel{n \rightarrow \infty}{\longrightarrow} n\pi_{\bullet}$$

Starting from  $a$  inside transient state  $\mathcal{T}$ :

Same as before for finite  $n$

The limit is a bit different:

- If the target state is transient

$$W_{ij}^{(\infty)} = \delta_{ij} + \sum_{k \in \mathcal{T}} P_{ik} W_{kj}^{(\infty)} \quad \text{for every } i \in \mathcal{T}$$

where  $\delta_{ij} = 1$  if  $i = j$  and 0 otherwise.

- If the target is in recurrent state then as before

Finally notice that

$$k_i = \sum_{j \in \mathcal{T}} W_{ij}$$

### Compute Mean and Variance of the first passage time to state $a$

Let  $k_i^a$  = mean time to reach  $a$  from  $i$

Do one system of this form for each required  $a$

Clearly  $k_a^a = 0$  however, for the final result consider  $k_a^a = 1/\pi_a$

$$k_i^a = 1 + \sum_{j \neq a} P_{ij} k_j^a$$

If  $i \nrightarrow a$ , or if  $i \rightarrow a$  but the chain can be trapped in another closed class disjoint from  $a$ , then  $k_i^a = \infty$

Let  $\alpha_i^a$  = secon moment of time to reach  $a$  from  $i$

Once the means  $k_i^a$  are known just build the following system

$$\alpha_i = 1 + \sum_{j \neq a} P_{ij}(2k_j + \alpha_j)$$

Finally the variance is

$$\alpha_i - k_i^2$$

**Computing  $\lim_{n \rightarrow \infty} P^n$ :**

- Doesn't exist for periodic classes "X"
- Is null for transitions (from transient) into transient states
- Doesn't exist for transitions (from transient) into periodic classes "X"

For Positive Recurrent Aperiodic Class it is just the steady state probability:

For a submatrix  $P_{C_i}$  where  $C_i$  is a pos. recurrent aperiodic class the rows will be

$$\text{row } i: (\pi_1 \quad \pi_2 \quad \dots \quad \pi_n)$$

For transitions (from transient) to pos recurrent aperiodic it is the abs probability times steady state of the state it transitions into

$$P(\text{absorbed in } C_i) \cdot \pi_n \quad \text{for all entries}$$

**Computing  $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n P^k$ :**

- Exists *always*
- If standard limit exists, it is equal to it

For Positive Recurrent Periodic Class it is just the steady state probability (same as before for aperiodic)

For For transitions (from transient) to pos recurrent periodic it is the abs probability times steady state of the state it transitions into (same as before fro transient  $\rightarrow$  aperiodic)

| Transition Type (i $\rightarrow$ j)                           | Class of i      | Class of j                         | Limit $\lim_{n \rightarrow \infty} P_{ijn}$                              |
|---------------------------------------------------------------|-----------------|------------------------------------|--------------------------------------------------------------------------|
| <b>To Transient</b>                                           | Any             | Transient                          | 0                                                                        |
| <b>Across Recurrent</b>                                       | Recurrent $C_1$ | Recurrent $C_2$ ( $C_1 \neq C_2$ ) | 0                                                                        |
| <b>Within Aperiodic Recurrent</b>                             | Recurrent $C_i$ | Same $C_i$ (Aperiodic)             | $\pi_j$ (from $\pi_{C_i} P_{C_i} = \pi_{C_i}, \sum \pi = 1$ )            |
| <b>Within Periodic Recurrent</b>                              | Recurrent $C_i$ | Same $C_i$ (Periodic)              | <b>Does not exist (X)</b>                                                |
| <b>Transient <math>\rightarrow</math> Aperiodic Recurrent</b> | Transient       | Recurrent $C_i$ (Aperiodic)        | $P(\text{absorbed in } C_i \mid X_0 = i) \cdot \pi_j$                    |
| <b>Transient <math>\rightarrow</math> Periodic Recurrent</b>  | Transient       | Recurrent $C_i$ (Periodic)         | 0 if unreachable; <b>Does not exist (X)</b> if reachable and oscillating |

### Mean recurrence time (avg return times)

For pos recurrent state  $\mu_i = 1/\pi_i$

For transient or neg rec states  $\mu_i = \infty$

## 5.4) First Step Analysis

General formula:

$$W_{ij} = \mathbb{I}\{i=j\} + \sum_k P_{ik} W_{kj}$$

This is used for

- avg visits
- passage times

## 5.5) GBN

If  $P$  is known, then also  $\pi_G, \pi_B$  are known by solving  $\pi = \pi P$  with  $\pi_G + \pi_B = 1$ .

If  $\pi_G, \pi_B$  is known  $P$  cannot be calculated. If also  $\mathbb{E}[\text{consecutive good/bad slots}]$  is known (one of the two) then

$$p_{GB} = \frac{1}{\mathbb{E}[\text{consec good}]} \quad p_{BG} = \frac{1}{\mathbb{E}[\text{consec bad}]}$$

From here a parametric  $P$  can be defined and by solving  $\pi = P\pi$  with  $a + b = 1$  as an additional condition the whole  $P$  is found.

If both are known it is directly found by recalling that each row = 1.

If the throughput of no protocol is asked:

$$\eta = \pi_G P[\text{success}|G] + \pi_B P[\text{success}|B]$$

A GBN on a two state MC works as follows. It has RTT of  $m$  slots

- Normal GBN with iid error  $\epsilon$  on forward channel and  $m$  slots to return

$$\eta = \frac{1 - \epsilon}{1 - \epsilon + m\epsilon}$$

(where  $P\{BG\} = 1 - \epsilon$  since iid and  $P\{GB\} = \epsilon$ )

- Error Free feedback throughput:

$$\eta = \frac{P_{10}^{(m)}}{P_{10}^{(m)} + mP_{01}} = \frac{P_{BG}^{(m)}}{P_{BG}^{(m)} + mP_{GB}}$$

- Feedback channel with iid error  $\delta$ :

$$\eta = \frac{(1 - \delta)P_{BG}^{(m)}}{(1 + (m - 1)\delta)P_{BG}^{(m)} + m((1 - \delta)P_{GB} + \delta P_{GB}^{(m)})}$$

If iid errors on both feedback and forward then use the last provided formula by recalling  $P_{BG}^{(2)} = P_{BG} = 1 - \epsilon$  since iid and  $P_{GB} = \epsilon$

## 5.6) Min/Max of two distributions

Let  $A, B$  be two non negative independent rvs, then define  $W = \min(A, B)$

$$W = \min(A, B) > t \iff A > t \cap B > t \longrightarrow P[W > t] = P[A > t]P[B > t]$$

Sometimes this is enough to know what the distribution is.

Calculate  $\mathbb{E}[W]$ :

$$\mathbb{E}[W] = \int_0^\infty P(W > t) dt$$

Recall also:

$$F_W(t) = 1 - (1 - F_A(t))(1 - F_B(t))$$

Specific cases:

- $A \sim \text{Exp}(\lambda_A)$   $B \sim \text{Exp}(\lambda_B)$ :

$$W \sim \text{Exp}(\lambda_A + \lambda_B)$$

$$\mathbb{E}[W] = \frac{1}{\lambda_A + \lambda_B}$$

- $A \sim \text{Exp}(\lambda)$   $B \sim \text{Constant}(T)$

$$W \sim \text{Truncated Exp}(\lambda) \text{ at } T$$

$$\mathbb{E}[W] = \frac{1 - e^{-\lambda T}}{\lambda}$$

In general a uniform rv truncates the other rv.

Now notice:

$$\max(A, B) + \min(A, B) = A + B \rightarrow \max(A, B) = A + B - \min(A, B)$$

Therefore the expectation of the max is very easily obtained:

$$\mathbb{E}[\max(A, B)] = \mathbb{E}[A] + \mathbb{E}[B] - \mathbb{E}[W]$$

So the distribution of max can be found by finding the distribution of min:

$$P[\max(A, B) > t] = P[A > t] + P[B > t] - P[\min(A, B) > t]$$

$$= P[A > t] + P[B > t] - P[A > t]P[B > t]$$